

Noah M Crabb

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EDUCATION

Virginia Polytechnic Institute, College of Engineering

Bachelor of Science: Aerospace and Ocean Engineering

Expected B.S Graduation: May 2027

GPA: 3.71/4.00 – *President's List Fall 2024, Dean's List Spring 2025*

Expected M.S Graduation: May 2028

Junior standing in an accelerated B.S./M.S. program (typically 5 years), on track to **complete both in 4 years**

Relevant Coursework: Astromechanics, Aerothermodynamics & Propulsion, Fluid Dynamics, Aerospace Structures, System Dynamics and Control, Computational Methods, Aerodynamics & Compressibility

WORK EXPERIENCE

Pi Sigma Epsilon – Delta Epsilon Chapter, Virginia Tech

Blacksburg, Virginia

Professional Member

Nov 2025 – Present

- Selected as **1 of 26 members from 120+ applicants** through a competitive, multi-week recruitment process including interviews, case studies, and professional evaluations.
- Completed an **intensive 8-week development program** with 4 meetings per week, focusing on communication, teamwork, project coordination, and business acumen—including **proficiency with Microsoft/Excel tools**.
- Synthesized **quantitative performance and financial results** into professional presentations, bridging engineering-style analysis with non-technical audiences.

3DPAC

Blacksburg, Virginia

CAD Lead / CAD Engineer

Jan 2025 – Present

- Promoted to CAD Lead in Mar. 2026** after serving as one of two CAD engineers responsible for full-aircraft CAD development; now oversee CAD members, delegate modeling tasks, track deadlines, lead Fall CAD recruitment, and coordinate design standards across the full team.
- Lead technical CAD meetings, member training, and formal design reviews** to teach Fusion360/SOLIDWORKS workflows, evaluate aircraft assemblies, resolve design and manufacturability issues, and keep competition deliverables on schedule.
- Co-developed complete competition aircraft CAD models and supported rapid design–build–test cycles using **Creality 3D printers**, contributing to a national **1st Place Most Innovative Design** award and **2nd Place Time Aloft** finish at UT Arlington's 3D Printed Aircraft Competition.

PROJECTS

Wind Tunnel Boundary Layer & Sting Mount Research Project

Blacksburg, Virginia

Experimental Aerodynamics Team Member

Apr 2026

- Conducted wind tunnel testing in the **Virginia Tech Stability Wind Tunnel**, measuring turbulent boundary layer velocity profiles with a 30-port pneumatic rake across multiple Reynolds number conditions
- Processed pressure data in **MATLAB** to compute normalized velocity profiles, local freestream velocity, and boundary layer thickness, comparing experimental results to turbulent boundary layer power-law predictions.
- Analyzed the effect of **sting mount geometry** on near-wall flow development, connecting boundary layer growth trends to adverse pressure gradients, horseshoe vortex behavior, aerodynamic blockage, and CFD validation.

Balsa Wood Glider Design Project

Blacksburg, Virginia

Glider Design & Analysis Team Member

Jan 2025 – May 2025

- Collaborated with a 4-member engineering team to design and theoretically prototype a hand-launched balsa wood glider optimized for maximum flight distance, balancing aerodynamic performance, stability, and structural feasibility.
- Conducted **MATLAB aerodynamic simulations**, CAD modeling, and theoretical stability analyses to refine airfoil geometry, improving lift-to-drag ratio and validating performance against expected metrics.
- Supported iterative design, manufacturing, and performance evaluation of a balsa wood glider, using analysis results to improve flight stability, glide efficiency, and overall distance potential.

OTHER

- Skills:** SOLIDWORKS, Fusion360, Python, MATLAB, Excel, Revit, XFLR5, Microsoft 365, AutoCAD, aircraft assembly design, aerodynamic analysis, boundary layer analysis, wind tunnel testing, 3D printing, rapid prototyping.
- Interests:** Pickleball, Soccer, CAD-Based Design Challenges, Plane spotting, Propulsion